

Sustainability university rankings: a comparative analysis of UI green metric and the times higher education world university rankings

Barbara Galleli, Noah Emanuel Brito Teles,
Joyce Aparecida Ramos dos Santos and
Mateus Santos Freitas-Martins

Department of Management, Federal University of Parana, Curitiba, Brazil, and

Flavio Hourneaux Junior
*School of Economics, Business and Accounting, University of Sao Paulo,
Sao Paulo, Brazil*

Abstract

Purpose – This study aims to answer the research question: How to evaluate the structure of global university sustainability rankings according to the Berlin Principles (BP) framework.

Design/methodology/approach – The authors investigated two global sustainability rankings in universities, The UI green metric World University Ranking (WUR) and the Times Higher Education World University Rankings (THE-WUR). The authors performed content analysis regarding their evaluation criteria and assessed both rankings using the BP framework.

Findings – Results show that there is still a gap to be filled regarding the specificity of global university sustainability rankings. Although the THE-WUR had a better performance in this research, there are several items for improvement, especially regarding the methodological procedures. There are structural differences, limitations and points for improvement in both rankings. Besides, it may not be possible to have a unique and more appropriate ranking, but one that can be more suitable for a contextual reality.

Practical implications – This study can be helpful for university managers when deliberating on the most appropriate ranking for their institutions and better preparing their higher education institutions for participating in sustainability-related rankings. Besides, it suggests possible improvements on the rankings' criteria.

Social implications – The authors shed light on challenges for improving the existing university sustainability rankings, besides generating insights for developing new ones. In a provocative but constructive perspective, the authors question their bases and understandings of being “the best university” regarding sustainability.

Originality/value – This is the first study that provides an in-depth analysis and comparison between two of the most important global university sustainability rankings.

Keywords Higher education institutions, Sustainability performance, Sustainability university rankings, Berlin principles

Paper type Research paper



Introduction

Higher education institutions (HEIs) occupy a central role in supporting the advancement of sustainable development (SD). Besides playing a crucial part in producing and disseminating knowledge, HEIs can promote the basis for learning toward the complexity of behavior and decisions required by SD, in a global perspective of future-oriented responsibility and a participatory and prejudice-free process (Barth *et al.*, 2007). HEIs can also build a fairer and more sustainable society, especially by educating future professionals and global citizens (Zamora-Polo *et al.*, 2019).

Themes related to SD can be found in HEIs' global agenda since the 1980s (Leal Filho, 2011). Due to numerous meetings and agreements spanning the 1990s and 2000s, their presence has not but grown. Therefore, HEIs have been challenged to implement sustainability in an integrated and comprehensive manner (Ramos *et al.*, 2015; Liu *et al.*, 2019) by covering a broad extension of activities, such as education, research, operation and campus experiences, outreach, institutional structure and evaluation and report (Lozano *et al.*, 2013).

Regarding the latter, evaluation and report practices, some HEIs networks for SD have championed the application and extension of university sustainability ranking systems, designed to assess universities' performance concerning practices related to environmental and social issues (Puertas and Martí, 2019). Although not being free from some criticism (Lauder *et al.*, 2015), the development of university sustainability ranking systems may help university managers focus their attention toward SD and speed up integrating sustainability into HEIs (Shi and Lai, 2013) and foster the idea of excellence (Taylor and Braddock, 2007).

Then, it is possible to identify several indices developed to quantify the contribution made by HEIs. Some of the most important are the sustainability tracking, assessment and rating system (STARS); the auditing instrument for sustainability in higher education (AISHE); the assessing responsibility in sustainable education (ARISE); the green league (now People and Planet's University League); the UI Green Metric World University Ranking (UI-GM); and the Times Higher Education World University Rankings (THE-WUR).

The assessment of sustainability in universities has been examined with several critical reviews and meta-analyses on using various assessment tools (Lauder *et al.*, 2015). For instance, Boer (2013) discussed the concept of education for SD (EdS) through the prism of different frameworks, namely, STARS, AISHE, ARISE and International Organization for Standardization methods. Suwartha and Sari (2013) evaluated one ranking result from its very first year of publication. Sonetti *et al.* (2016) compared universities' performance based on their sustainability scores. Finally, Draheim *et al.* (2019) applied the indices in institutes of technology and polytechnic universities. Other authors, such as Lozano (2006) and Puertas and Martí (2019), developed alternative sustainability indexes from the variables used in previous tools while Shi and Lai (2013) proposed a new assessment framework.

We can notice that literature has provided a great deal of research on sustainability in higher education and tools for the assessment, measurement and rankings. However, most of these studies have reported on regional, national or local contexts or are case studies of a single university's attempts to establish and measure sustainability. There is still relatively little in the literature on global sustainability rankings in higher education (Lauder *et al.*, 2015). Besides, there is a need for a more in-depth analysis of sustainability-related rankings to provide knowledge for implementing sustainability practices and developing appropriate policies in HEIs (Lazzarini and Pérez-Foguet, 2018; Shi and Lai, 2013).

The literature also points out that there are several gaps to fill and opportunities to improve university sustainability rankings. There are advances to be made on establishing frameworks and prioritizing proper sustainability concepts that can act as mechanisms or principles for designing methods and selecting categories and indicators (Lauder *et al.*, 2015). There is also a claim for developing or reviewing indicators that evaluate universities from more objective perspectives and stimulate universities to introduce, register and publish the information in this field (Luckman *et al.*, 2010).

Given this scenario, we propose the following research question: how to evaluate the structure of global university sustainability rankings according to the Berlin Principles (BP) framework? Then, we analyzed and compared two global university sustainability rankings, the UI-GM and the THE-WUR, for their contemporaneity, classification of institutions and international application. As argued by Diez-Cañamero *et al.* (2020), the measuring of sustainability in organizations presents a positive evolution and must continue to grow. However, at the same time, it is surrounded by a broad and heterogeneous universe that provokes confusion and dilemmas when selecting the most proper tool to evaluate sustainability performance or making decisions based on indexes, ratings and rankings' results. Hence, identifying the common and different characteristics of such tools, i.e. comparing them qualitatively, contributes to putting this chaotic universe in order.

Regarding academic contributions, it is expected that the results of this study can contribute to the knowledge construction on sustainability field by suggesting new research questions concerning sustainability university rankings and critically analyzing the topic of sustainability measurement in this context. To the best of our knowledge, this is the first study that analyzes the THE-WUR. From the insights of this study, we intend to contribute to the scarce literature on global sustainability rankings in higher education (Lauder *et al.*, 2015). Besides, this paper is intended to facilitate researchers' work and the academic community by exposing similarities and differences of the two single global sustainability university rankings.

This paper is structured as follows: in the next section, we present a brief literature review on university sustainability rankings; we describe the research method used and subsequently present and discuss the results. The paper ends with conclusions, implications, limitations and suggestions for future studies.

Sustainability in higher education institutions

Environmental problems and the search for solutions to combat climate change (Suwartha and Sari, 2013) and problems related to poverty and violence (Albareda-Tiana *et al.*, 2018) have become a matter of interest for university leaders. From acknowledging this relevance, sustainability initiatives and practices have started to be globally implemented by universities worldwide (Trowler *et al.*, 2013; Zutshi *et al.*, 2019).

The role of universities in collaborating for sustainability lies in guiding future leaders, professionals and citizen to an education based on the principles of sustainability, preparing them to implement expressive changes in response to current and future crises (Hooey *et al.*, 2017; Storey *et al.*, 2017; Zamora-Polo *et al.*, 2019). The HEIs are responsible for disseminating knowledge and instigating behavior to build a society that understands and acts toward sustainability, contributing through its practices to the well-being of future generations (Blanco-Portela *et al.*, 2018; Farinha *et al.*, 2019).

To be considered a sustainability-oriented institution, universities have the challenge to holistically integrate practices involving social, environmental and economic dimensions throughout their operations, structures, teaching, research and outreach. Fragmented initiatives involving isolated actions by specific university groups or departments are not

enough; sustainability must be institutionalized at all campus levels (Blanco-Portela *et al.*, 2018; Hooey *et al.*, 2017).

Lozano *et al.* (2015) add that sustainable universities consistently act with sustainability practices in education, research, extension, campus management and institutional policies and experiences. Initiatives are observed at various domains, such as in the institutional structure from the development of sustainability policies, insertion of sustainability in the mission, vision of the university, creation of specific departments and signing of declarations related to the cause.

Literature refers to a diversity of sustainable practices observed on the campus, such as environmental management planning for the implementation of sustainable buildings and the sharing of information from sustainability reports with indicators on the use of water, energy, emissions, waste, transportation (Hooey *et al.*, 2017).

Practices associated with EdS are also part of the framework of sustainability-oriented institutions. These involve teaching and learning practices (Butt *et al.*, 2014), courses and programs focused on teaching sustainability proposed to educate teachers about the topic (Lozano *et al.*, 2015).

University sustainability rankings

HEI rankings are increasingly common tools (Razak *et al.*, 2013), although their relevance and practical implication in society were not met with consensus. Rankings, in general, can be understood as classifications of universities or departments of a given area of knowledge, in numerical order, based on a set of predefined criteria (Webster, 1986; Marginson and Van Der Wende, 2007).

In assessing issues such as research, academic reputation, quality of education (Lukman *et al.*, 2010; Ragazzi and Ghidini, 2017) and sustainability-related performance in more specific cases (Suwartha and Sari, 2013), such instruments can be considered beneficial to HEIs. Besides being capable of demonstrating a parameter of undertaken activities, by evincing strengths and points for improvement, these rankings can contribute to the legitimation of an HEI to its stakeholders (Huang, 2011). The impact of HEIs' activities on sustainability is increasingly being charged by the stakeholders of these institutions (Findler *et al.*, 2019).

The recognition of HEIs as protagonists in leading societies toward an economically balanced, socially fair and environmentally responsible future (Vargas *et al.*, 2019) has resulted in the development of several university sustainability rankings. These instruments are intended to assess the reputation of universities and their performance toward environmental and social concerns (Puertas and Marti, 2019), such as environmental improvements made on their campi, the inclusion of sustainability courses in curricula and the positioning of universities toward workers' rights (Cotton *et al.*, 2018).

Sustainability-related university rankings are essential to direct the focus of university managers on the importance of SD actions, assisting in the institutionalization of sustainability (Shi and Lai, 2013), as they may provide the alignment of sustainability practices with institutional goals (Lazzarini and Pérez-Foguet, 2018). These tools are relevant when one observes that if, on the one hand, universities are emphatically and traditionally compared from an educational and research perspective and on the other, social and environmental issues have received little or no attention in conventional rankings. In this context, universities' efforts toward sustainability can end up not receiving proper recognition (Lukman *et al.*, 2010).

Despite the widespread dissemination of university rankings, a common criticism is related to the non-systematicity of categories, indicators and methodologies present in these

tools. The literature points out that it is often unclear, for instance, why to choose a specific evaluation methodology or indicator, how to check its validity, who was responsible for the criteria decisions and how open and reflective the decision process was (Lauder *et al.*, 2015; Lukman *et al.*, 2010).

From an outsider's point of view, the consequence is that the information provided is often challenging to decipher. In addition to being conveyed in somewhat "nebulous" academic language, the performance results are presented in a way that does not facilitate comparison among institutions, especially considering their different national contexts (Hazelkorn, 2014). Thus, in the case of global rankings, the methodological discrepancy becomes both more transparent and more severe because, together with the variability of definitions and criteria, there are also the specificities of contexts that permeate countries with different levels of development (Razak *et al.*, 2013).

Beyond the challenges of the traditional rankings, regarding the sustainability ones, there is an extra effort to join universities in countries where SD issues are not essential elements for most people – commonly in developing countries. The challenge is to demonstrate the potential benefits of institutions' participation in such instruments. It is also apparent that different sustainability issues captured in the choice of categories, indicators and weightings are not equally relevant in all countries (Lauder *et al.*, 2015).

Another criticism is that using rankings as parameters to measure sustainability practices in HEIs may be contradictory. Davey (2017) argues that with the use of standardized frameworks for university sustainability assessment, the HEI becomes a collector and reporter of data rather than a *locus* of discussion of how to define, execute and measure sustainability practices. In this sense, the rank evaluation may induce the standardization of practices by universities (Bizzerril, 2018) and even if many HEIs have committees, reports, offices, they do not necessarily change the behavior at the campus (Luckman *et al.*, 2010).

Method

To answer the research questions and accomplish the study's aim, we analyzed two ranking systems as follows: the UI-GM and THE-WUR. They were chosen for their contemporaneity and holistic classification of institutions and international application.

The UI-GM is organized by Universitas Indonesia and is considered the first attempt to make a global ranking of universities' sustainable behavior (Ragazzi and Ghidini, 2017). It aims to assess policies and activities within green campuses to promote a sustainability culture in HEIs. This ranking is suitable for universities in both developed and developing countries and is, therefore, considered a global ranking (Suwartha and Sari, 2013). The ranking is also seen as the most important global sustainability ranking for universities (Ragazzi and Ghidini, 2017).

The THE-WUR published its first edition in 2019. It assesses universities' impacts against the United Nations' Sustainable Development Goals (SDGs). The THE-WUR is published by TES Global and affiliated with The Times, a recognized British newspaper. It is one of the rankings modalities published by the THE, since 2004. According to the World Economic Forum, this is the world's first global attempt to document evidence of universities' impact on society, beyond the traditional research and teaching performance (Bothwell, 2019b).

The data were collected through secondary sources, like the information provided on the rankings' websites and their own guidelines (UI Green Metric World University Rankings, 2019a, 2019b; THE World University Rankings, 2019a, 2019b). Most of the information provided by the rankings' websites was from the year 2019.

Besides a comparative content analysis and based on previous studies (Lauder *et al.*, 2015; Ragazzi and Ghidini, 2017; Shi and Lai, 2013; Shriberg, 2002; Stolz *et al.*, 2010), we evaluated both ranking systems using the BP framework, which provide quality and good practices in HEI rankings. The BP on Ranking of HEI was declared during a meeting in the city of Berlin in 2006 by the International Ranking Expert Group (IREG), founded by the United Nations Educational, Scientific and Cultural Organization European Centre for Higher Education. The goal was to establish a set of principles of quality and good practice in HEI rankings, which were becoming a global phenomenon (IREG, 2006). To that end, we made use of the scales of assessment proposed by Stolz *et al.* (2010).

To avoid possible bias in the analysis of the criteria covered by the BP, three researchers participated in this analysis. Objective, impartial and consensual discussions led to the results found. The steps of data collection, synthesis and analysis were carried out over two months, from November to December 2019.

Results

First, in Table 1, we present the comparative content analysis of the two rankings, UI-GM and THE-WUR. For that, 11 dimensions were chosen, based on the ranking's configurations and previous similar studies (Lauder *et al.*, 2015; Ragazzi and Ghidini, 2017). Most of the categories were also used when Diez-Cañamero *et al.* (2020) qualitatively compared corporate sustainability systems.

Regarding the year of creation, Table 1 shows the first edition of UI-GM was made available in 2010 while the THE-WUR, had its first edition in 2019. Although the number of HEIs participants on the last published edition on the UI-GM (780) is almost double of participants of the THE-WUR (462), the number of countries is not discrepant, 85 for the former and 76 for the latter. Considering the countries most represented in the ranking, it is notable that in UI-GM there is a prevalence of Eastern countries while for the THE-WUR there is a balance, with a greater presence of Western countries. We see that Russia appears in both as one of the most represented countries.

Concerning the expressed objectives, it should be noted that both rankings seem to have distinct but complementary purposes. While the UI-GM seeks to provide a parameter on sustainable conditions and sustainability policies in institutions, THE-WUR seems to transcend that purpose by measuring how successful these institutions are toward meeting SDGs. Parallel to this, the adherence of participating institutions can also be considered diverse in these instruments. The reason for this is because while in the UI-GM the universities, besides being invited, HEIs can send their information voluntarily, in THE-WUR only the second modality is adopted.

From the information requested and criteria, we can observe the most remarkable difference of focus from the UI-GM and the THE-WUR. The former requires data related to university size, campus location, information on energy use, water and waste disposal, as well as transportation and energy use, that will be analyzed through six categories, associated to the campus structure (Setting and Infrastructure and Transport), the institution's missions (Education and Research) and environment (Energy and Climate Change; Waste; Water). The latter, in 2019, used the SDGs frameworks as a parameter, thus seeking information on how universities are committing to 11 out of the 17 SDGs. For 2020, it is known that the whole set of objectives will be considered (THE University-IRs 2019). While the UI-GM demands data in all the categories, THE International Ranking (IR) posits that at least four of the SGD should be necessarily covered as follows: the "SDG 17 – Partnership for the goals" and at least three others for the university be part of the overall ranking.

Table 1.
Comparative content
analysis of UI-GM
and the university
impact ranking

Dimension	IU GM	THE university impact ranking
Year of creation	First edition in 2010	First edition in 2019
Total of HEI participants (2019)	780	462
Total of countries participants (2019)	85	76
Countries most represented in the ranking (2019)	Indonesia (72 HEIs); Russia (46 HEIs); Iraq (44 HEIs); Turkey (43 HEIs); Pakistan (40 HEIs) (UI Green Metric World University Rankings, 2019a)	Japan (41 HEIs); USA (31 HEIs); Russia (30 HEIs); UK (26 HEIs); Canada (25 HEIs) (THE, 2019 b)
Ranking objective	The aim of this ranking is to provide the result of the online survey regarding the current condition and policies related to Green Campus and Sustainability in Universities all over the world (UI Green Metric World University Rankings, 2019a, 2019b)	The aim of the ranking is to document the impact of universities on society, rather than just research and teaching performance (Bothwell, 2019a). The ranking measures global universities' success in delivering the UN SDGs (THE, 2019a)
Form of participation	Universities around the world that already have a presence in other World Universities Rankings are invited to take part. Additionally, universities can submit data by filling in our online survey form (UI Green Metric World University Rankings, 2019a, 2019b)	Universities can submit data on as many of the 11 SDGs considered for the ranking, as they are able. Any university that provides data on SDG 17 and at least three other SDGs is included in the overall ranking (THE, 2019a)
Information requested	The information requested includes data on the size of the university, both spatially and in terms of population, the campus location and the amount of green space; and information on energy use, transport, water use and recycling and waste treatment. In addition, the ranking asks about efforts being made by the institution toward establishing green policies and management (UI Green Metric World University Rankings, 2019a, 2019b)	Information on how HEIs are committed to 11 of the 17 SDGs (THE, 2019a)
Criteria	Setting and infrastructure; energy and climate change; waste; water; transportation; education and research (UI Green Metric World University Rankings, 2019a, 2019b)	SDG 3: good health and well-being; SDG 4: quality education; SDG 5: gender equality; SDG 8: decent work and economic growth; SDG 9: Industry, innovation and infrastructure; SDG 10: reduced inequalities; SDG 11: sustainable cities and communities; SDG 12: responsible consumption and production; SDG 13: climate action; SDG 16: peace, justice and strong institutions; SDG 17: partnerships for the goals (UN, 2015)
Evaluation methodology	The evaluation instruments are based on a broad philosophy that encompasses the environment, economics and equity. Scoring for each item is numeric so that the data can be processed statistically.	Calibrated indicators are used to provide comprehensive and balanced comparisons across three broad areas: research, outreach and stewardship. Each SDG has a series of metrics that are

(continued)

Dimension	IU GM	THE university impact ranking
	Scores are simple counts of things or responses on a scale of some sort. Each of the criteria is categorized in a general class of information and when the results are processed, the raw scores are weighted to give a final calculation. The weights are as follows: setting and infrastructure 15%; energy and climate change 21%; waste 18%; water 10%; transportation 18%; education 18%. The criteria and the weightings are continuously reviewed to reflect input from participants and state of the art developments in the field (UI Green Metric World University Rankings, 2019a, 2019b)	used to evaluate the performance of the university in that SDG. The score from each SDG is scaled so that the highest score in each SDG in the overall calculation is 100. This is to adjust for minor differences in the scoring range in each SDG and to ensure that universities are treated equitably whichever SDGs they have provided data for. A university's final score in the overall table is calculated by combining its score in SDG 17 with its top three scores out of the remaining 10 SDGs. SDG 17 accounts for 22% of the overall score while the other SDGs each carry a weighting of 26% (THE, 2019a)
Evidence requests	Mandatory; lack of evidence decreases the score. Evidence can be photos, charts, tables and other data (UI Green Metric World University Rankings, 2019a, 2019b)	Mandatory; Evidence related to each indicator is requested from universities and are subsequently evaluated, seeking cross-reference of information when there is uncertainty (THE, 2019a)
Time for analysis of HEI's data (2019)	Annually, between May – October (UI Green Metric World University Rankings, 2019a)	Past 5 years (2013–2018) (THE, 2019a)

Source: The authors

Table 1.

Concerning the evaluation methodology, the UI-GM declares that it was not based on any existing ranking system, but developed with an awareness of several current sustainability assessment systems and academic university rankings, including the THE-WUR, STARS and Green Report Card. Each of the six criteria is weighted and has its own indicators (four to eight per category). It is claimed that the design of UI-GM was informed by experts in both rankings and in sustainability. All the criteria are individually categorized in a general class of information and when the results are processed, the raw scores are weighted to give a final calculation ([UI Green Metric World University Rankings, 2019b](#)).

For the THE-WUR, the evaluation methodology is declared to be developed in conjunction with THE's partners, Vertigo Ventures and Elsevier and after consultation and input from individual universities, academics and sector groups. A university's final score in the overall table is calculated by combining its score in SDG 17, weighting 22%, with its top three scores out of the remaining 10 SDGs, with a 26% weight each. This fact means that different universities are scored based on a different set of SDGs, depending on their focus – or their data provision. Data will come from a variety of sources, including direct submissions from institutions and bibliometric data sets from Elsevier ([THE World University Rankings, 2019c](#)).

Finally, regarding the requested evidence, both rankings require universities to provide evidence of actions related to the indicators. On their websites, both provide examples of the necessary evidence. For the UI-GM, the lack of evidence causes a decrease in the score ([UI Green Metric World University Rankings, 2019b](#)). THE IR analyzes the evidence seeking to crosscheck information when there is uncertainty about the evidence provided. If the

university has no evidence on the requested data, it should not be provided then (THE World University Rankings, 2019c).

Given the content analysis, we evaluated both ranking systems using the BP framework, from the scales of assessment proposed by Stolz *et al.* (2010). We followed the authors' recommendations for the analysis of each principle and scale scoring. We also agreed with the author's exclusion of six BPs, considering the purpose of this study. Table 2 presents such analysis.

As we can see in Table 2, the THE-WUR performed higher in four of the 17 principles analyzed comparing to UI-GM. These principles are in the dimensions of evaluation methodology (4 A; 8) and consumer friendliness (3; 15B). In the principles that make up the transparency dimension, both rankings scored equally.

BP 3 is the first of the BP's criteria in which there was a difference among the rankings. It is related to the classifications and/or scores that consider the specificities of the participating HEIs, as their mission and goals. While UI-GM does not have any campus specific ratings, THE IR does rank according to the HEI's program area. The other principle that presented scoring divergence is BP 4 A, which deals with the sources of information used to classify HEIs. While in UI-GM, universities themselves provide information by completing an online form, in THE-WUR, besides the online form completed by the university, at least two bibliometric metrics are used and differ for each SDG (THE World University Rankings, 2019a).

In BP 8, which concerns the framing of the outcomes evaluated in terms of inputs or outcome measures, the ranking criteria showed that while in the UI-GM nearly 30% of the indicator measures are outcomes, in THE-WUR 50% make up this category. Finally, in BP 15B while the UI-GM does not allow any changes on the weights assigned to indicators, the THE-WUR allows the HEIs to modify the weights on the SDGs. The THE-WUR allows universities to be ranked based on a different set of SDGs, depending on their focus and their choice of disclosure.

Figure 1 presents the results of the comparative performance process. As suggested by Stolz *et al.* (2010), a final score, as well as separate sub-scores for methodology, transparency and consumer friendliness, were computed. A score of 5 indicates an excellent congruence between the ranking practices of the two rankings in analysis and the best practices proposed by the BPs; a score of 4 is interpreted as indicating good congruence, a score of 3 as fair congruence, a score of 2 as poor and a score of 1 as no congruence. Based on the BP, both rankings were above average and THE-WUR presented a higher final comparative performance than UI-GM.

Figure 1 shows the performance for UI-GM (dark blue dashed line) and for THE-WUR (light blue line) in each one of the 17 BP's criteria, split into the following three dimensions: transparency, methodology and consumer friendliness.

Discussion and implications

As the first point of discussion, we draw attention to the volume of HEIs participants on both rankings and their origin. The fact that the last edition of UI-GM had almost double the number of participants on THE IR is probably a reflection of the nearly 10-year tradition of the former and, as it is considered the very first sustainability-related global ranking for universities (Ragazzi and Ghidini, 2017). Regarding the representativeness of countries, the prevalence of Eastern countries in UI-GM and the relative more significant presence of Western countries in the THE-WUR is possible related to the origin of the maintainer organization, Indonesia and UK, respectively.

BP	Scale (Stolz et al., 2010)	GM	THE impact ranking
2	Be clear about their purpose and their target groups. Rankings have to be designed with due regard to their purpose. Indicators designed to meet a particular objective or to inform one target group may not be adequate for different purposes or target groups 2A 5 – Information is given about all of the following: target group; intended impact on the target group; scope of information provided to the target group; the impact that information provided to target group might have beyond direct ranking system-target group interaction 4 – Information is given about three of the above; 3 – Two of the above; 2 – One of the above; 1 – None of the above 2B 5 – All of the following dispositions can be found: data directly collected from the target group (i.e. through surveying) is used when computing an overall ranking score; the distinct message that this target group data reveals is identifiable when an overall ranking result is presented; a score of at least 4 has been achieved for Principle 3; an overall score of at least 4 has been achieved for Principle 15; 4 – Three of the above dispositions can be found; 3 – Two of the above; 2 – One of the above; 1 – None of the above 3 Recognize the diversity of institutions and take the different missions and goals of institutions into account. Quality measures for research-oriented institutions, for example, are quite different from those that are appropriate for institutions that provide broad access to underserved communities. Institutions that are being ranked and the experts that inform the ranking process should be consulted often	5 – Information is given about all the topics (UI Green Metric 2019a, 2019b) 3 – Data directly collected from the target group (UI Green Metric, 2019b) 1 – There are only separated scores for campus setting (UI Green Metric, 2019b)	5 – Information is given about all the topics (THE, 2019a, 2019b) 3 – Data directly collected from the target group (THE, 2019c) 2 – There are only separated scores by program area (THE, 2019c)

(continued)

Comparative analysis of UI green metric

Table 2.
Ranking's analysis based on the BP

Table 2.

BP	Scale (Stolz <i>et al.</i> , 2010)	GM	THE impact ranking
4	Provide clarity about the range of information sources for rankings and the message each source generates. The relevance of ranking results depends on the audiences receiving the information and the sources of that information (such as databases, students, professors, employers). Good practice would be to combine the different perspectives provided by those sources to get a more complete view of each HEI included in the ranking		
4A	5 – All four of the following sources of information used in the data collection process: Bibliometrics, students, professors, employers; 4 – Three of the above used; 3 – Two of the above; 2 – One of the above; 1 – None of the above/remains unclear	1 – The sources of information from the universities are unclear	2 – The sources of information concerning Bibliometrics used in the data collection process are clear (THE, 2019a) 1 – There is no clear information
4B	5 – The particular messages generated by all four different sources of information mentioned for 4 A are made explicit in the ranking 4 – The messages of three of the sources mentioned for 4 A are made explicit; 3 – Two; 2 – One; 1 – Messages remain unclear	1 – There is no clear information	
6	Be transparent regarding the methodology used for creating the rankings. The choice of methods used to prepare rankings should be clear and unambiguous. This transparency should include the calculation of indicators, as well as the origin of data		
6A	5 – Methodology given allows for recalculation of overall score on part of consumer 4 – All of the following methodological issues explained: the entity that has been measured for assessing particular indicator; the process of aggregating an indicator score; the process of weighting indicator scores to calculate ranking score 3 – Two of the methodological issues explained (as outlined for a score of 4); 2 – One of the above as outlined for a score of 4; 1 – None of the above as outlined for a score of 4	5 – Methodology given and explained (UI Green Metric, 2019b)	5 – Methodology given and explained (THE, 2019a)
6B	5 – Information given about origin of data for all indicators used 4 – For at least 75% of indicators used; 3 – For at least 50%; 2 – For at least 25%; 1 – for less than 25%, no	5 – Information is given about the origin of data for all indicators used are mandatory (UI Green Metric, 2019b)	5 – Information is given about origin of data for all indicators used are (continued)

BP	Scale (Stolz <i>et al.</i> , 2010)	GM	THE impact ranking
7	Choose indicators according to their relevance and validity. The choice of data should be grounded in recognition of the ability of each measure to represent quality and academic and institutional strengths and not the availability of data. Be clear about why measures were included and what they are meant to represent	assessment possible because issue not addressed by the ranking author(s) 5 – At least one indicator for all five dimensions: funding; human resources, academic outcomes; third mission and governance 4 – At least one indicator for four of the five dimensions; 3 – Three of the five dimensions; 2 – Two of the five dimensions; 1 – One of the five dimensions 7A 5 – All indicators are valid 4 – at least 75 % are valid; 3 – at least 50%; 2 – at least 25%; 1 – less than 25% or no assessment possible because issue not addressed by the ranking author(s) 7B 5 – 100% of the overall score calculated on basis of output measures 4 – At least 75% of the overall score; 3 – at least 50%; 2 – at least 25%; 1 – less than 25%, no assessment possible because issue not addressed by the ranking author(s)	mandatory (THE, 2019c) 5 – Indicators in all five dimensions are considered (UI Green Metric, 2019b) 1 – No assessment IS possible because the issue was not addressed by ranking author(s) 3 – 50% (THE, 2019a, 2019c)
8	Measure outcomes in preference to inputs whenever possible. Data on inputs are relevant as they reflect the general condition of a given establishment and are more frequently available. Measures of outcomes provide a more accurate assessment of the standing and/or quality of a given institution or program and compilers of rankings should ensure that an appropriate balance is achieved	5 – 100% of the overall score calculated on basis of output measures 4 – At least 75% of the overall score; 3 – at least 50%; 2 – at least 25%; 1 – less than 25%, no assessment possible because issue not addressed by the ranking author(s) 8 5 – 100% of the overall score calculated on basis of output measures 4 – At least 75% of the overall score; 3 – at least 50%; 2 – at least 25%; 1 – less than 25%, no assessment possible because issue not addressed by the ranking author(s)	1 – No assessment IS possible because the issue was not addressed by ranking author(s) 3 – 50% (THE, 2019a, 2019c)
9	Make the weights assigned to different indicators (if used) prominent and limit changes to them Changes in weights make it difficult for consumers to discern whether an institution or program's status changed in the rankings	5 – All weights assigned to the indicators used in the ranking are clearly stated 4 – Weights assigned are stated clearly for all sets of indicators used (i.e. category), but not for each indicator individually 3 – For at least 75% of sets of indicators (i.e. category) and/or indicators used; 2 – for at least 50%; 1 – for 9A 5 – All weights assigned to the indicators used in the ranking are clearly stated (UI Green Metric, 2019b)	5 – All weights assigned to the indicators used in the ranking are clearly stated (THE, 2019a, 2019c)

(continued)

Table 2.

Table 2.

BP	Scale (Stolz <i>et al.</i> , 2010)	GM	THE impact ranking
	due to an inherent difference or due to a methodological change		
		less than 50%, no assessment possible because issue not addressed by the ranking author(s)	
	9B	5 – No indicators used have been changed relative to the ranking most recently published 4 – Less than 5% of indicators used; 3 – less than 10%; 2 – less than 15%; 1 – more than 15%	5 – No changes were made on the indicators (UI Green Metric, 2019b, 2019c)
11	Use audited and verifiable data whenever possible. Such data have several advantages, including the fact that they have been accepted by institutions and that they are comparable and compatible across institutions	5 – 100% of the overall score calculated on basis of data gathered by third party institutions (collected on all universities included in the ranking) 4 – At least 75% of the overall score; 3 – At least 50%; 2 – At least 25%; 1 – Less than 25%, no assessment possible because issue not addressed by the ranking author(s)	1 – No assessment possible because issue not addressed by the ranking author(s)
12	Include data that are collected with proper procedures for scientific data collection. Data collected from an unrepresentative or skewed subset of students, faculty or other parties may not accurately represent an institution or program and should be excluded	5 – Description of data sampling procedures given for all indicators included in the ranking 4 – Given for at least 75% of indicators; 3 – For at least 50%; 2 – For at least 25%; 1 – For less than 25%, no assessment possible because issue not addressed by the ranking author(s)	1 – No assessment possible because issue not addressed by the ranking author(s)
		5 – Sampling procedure sound for all indicators included in the ranking 4 – For at least 75% of indicators; 3 – For at least 50%; 2 – For at least 25%; 1 – For less than 25%, no assessment possible because issue not addressed by the ranking author(s)	1 – No assessment possible because issue not addressed by the ranking author(s)
15	Provide consumers with a clear understanding of all the factors used to	A 5 – Definitions for all indicators used in the ranking are given 4 – For at least 75% of	5 – Definitions for all indicators (continued)

BP	Scale (Stolz <i>et al.</i> , 2010)	GM	THE impact ranking
develop a ranking and offer them a choice in how rankings are displayed. This way, the users of rankings would have a better understanding of the indicators that are used to rank institutions or programs. In addition, they should have some opportunity to make their own decisions about how these indicators should be weighted	indicators; 3 – For at least 50%; 2 – For at least 25%; 1 – For less than 25%, no assessment possible because issue not addressed by the ranking author(s) 15B 5 – Consumer can alter both weights assigned to indicators and the selection of indicators used in computing the ranking results for all of the following three levels: institutional, departmental, program 4 – alter both weights assigned to indicators and the selection of indicators for two of the three levels; 3 – for one of the three levels 2 – Consumer can alter either weight assigned to indicators or the selection of indicators used in computing the ranking results for at least one of the following three levels: institutional, departmental, program 1 – Consumer has no choice in how ranking is displayed	5 – Definitions for all indicators used in the ranking are given (UI Green Metric, 2019c) 1 – Consumer has no choice in how ranking is displayed (UI Green Metric, 2019c)	used in the ranking are given (THE, 2019c) 2 – Consumer can alter both weights assigned to indicators and the indicators (THE, 2019c)

Source: The authors, based on Stolz *et al.* (2010)

Table 2.

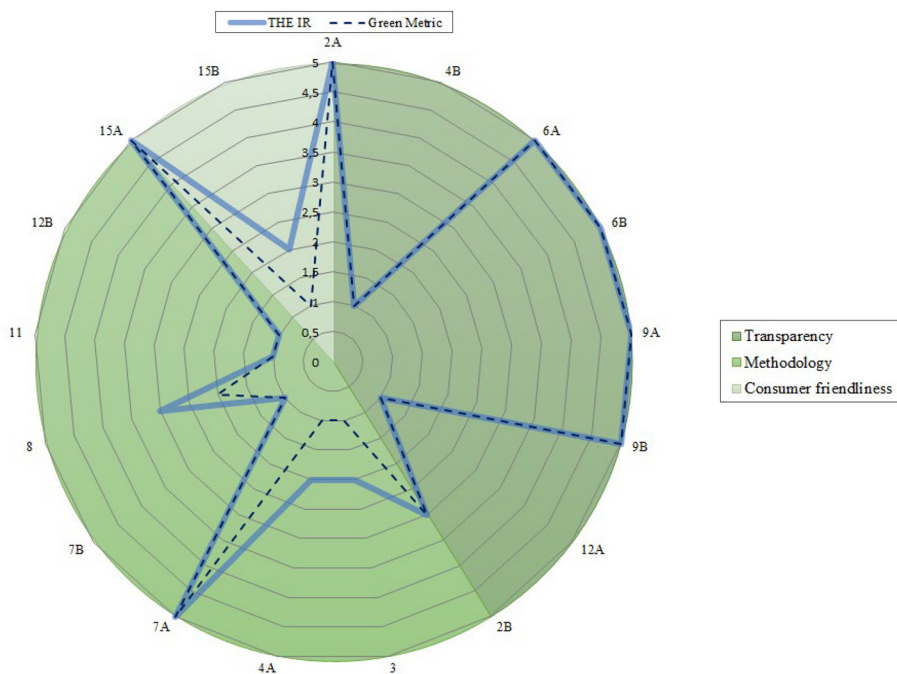


Figure 1.
Scores for UI-GM and
THE IR, based on the
BP

Source: The author, based on stolz *et al.* (2010)

Turning to the content analysis of the rankings, we note that UI-GM emphasizes more environmental aspects, with criteria that value how universities use natural resources, dispose of waste, provide/make use of green space on campus, etc. Such valuation of environmental dimension is evident when it comes to the weights for its criteria that together comprise 67% of the overall score. Meanwhile, although the THE-WUR also embraces environmental issues, it is also clear that it involves many more social problems, such as health and well-being, gender equality and reduction of inequalities. Even this outcome can be expected given that this ranking has the SDGs as background. What is more remarkable is that the indicators for each criterion related to the environment (SDGs 6, 7, 13, 14 and 15) prioritize research and education on the subjects rather than operational arrangements.

Nevertheless, it should be mentioned that while considerations on environmental and social criteria highlight the involvement of universities with issues that go beyond education and research, it is essential to note that the analysis of these criteria should not be seen as a guarantee of effective behavioral change on campus (Luckman *et al.*, 2010). This is especially true when one considers the institutions themselves provide all the information on actions in these spheres in both rankings.

Before forwarding the debate on which one is the best approach or focus for sustainability university rankings – if there is such a thing – the findings recall an extensively discussed issue in the area, which is the concept of sustainability itself (Vos, 2007). In none of the ranking documents or reports, it was possible to find a clear definition of sustainability or consider or identify a sustainable university. As Lauder *et al.* (2015)

claimed, first, any adequate definition of a complex concept should be constrained in some way to give the rankings clarity. More than that, a clear concept allows institutions to better fit their participation on the ranking and adequately positioning themselves concerning what they understand, privilege and disseminated concerning sustainability.

It is also essential to highlight the emphasis given to education. The UI-GM has a specific category for education, as does the THE-WUR. The difference in the indicators in this category in the UI-GM measures the number of courses, environmental and sustainability publications. At the same time, the THE-WUR is more concerned with lifelong learning opportunities provided, the number of graduates who gained primary school teaching qualifications and the proportion of first-generation students. This circumstance is evidence of the SDG framework, chosen by THE-WUR, in which the impact on society and the outcome of the university's mission matters more than merely input issues.

While, on the one hand, the framework provided by Agenda 2030 concerning SDGs, emphasizes and contributes to the ownership of a global agenda by HEIs, on the other hand, it may already determine a limit of applicability for THE-WUR, at least in the format in which it is presented today. In other words, in the way it is elaborated and presented, the THE-WUR would be finalized in 2030. There are also discussions on the difficulty of measuring the contributions of HEIs to SD (Leal Filho, 2011) and doubts about the possibilities of fulfilling this agenda in the face of the pandemic scenario caused by COVID-19 (Leal Filho *et al.*, 2020).

Regarding the comparison of rankings based on BP, initially, it should be noted that although UI-GM is older in editing, its methodological questions had lower scores than the THE-WUR in three (3; 4 A; 8) of the six principles analyzed in this category. We believe that the fact that the THE_WUR is linked to a traditional publisher (Times) and it has been involved in university rankings since 2004 may have put it ahead of UI-GM, even though it scored significantly below the maximum in these principles. This finding corroborates the criticisms regarding the methodological weaknesses that permeate university rankings (Lauder *et al.*, 2015. Lukman *et al.*, 2010).

In terms of transparency, both rankings scored equally on all six principles in this category, of which four achieved maximum score (2A; 6A; 6B; 9A) and two a minimum score (4B; 12A). On the one hand, the rankings have shown satisfactory results about aspects of clarity concerning the target group, methodology, the origin of the data/indicators used and weights defined in the rankings, confronting the argument that they convey information in a language sometimes tricky for outsiders to understand (Hazelkorn, 2014). An opportunity for improvement is related to transparency in the description of sampling procedures and the source of data.

As for the component results of the consumer-friendliness dimension, the THE-WUR scored better. Regarding the straightforward availability of definitions for all ranking indicators, both obtained a maximum score (15A). However, concerning offering consumers the choice of altering both weights assigned to the indicators and selecting the indicators used in computing the ranking results (15B), both rankings scored way below the maximum score. While UI-GM does not provide this possibility, the THE-WUR allows for a combination of the obtained score, allowing HEIs to be ranked based on different SDGs' performance combinations, depending on their focus and disclosure. Criticism about how detrimental rankings can be – due to the establishment of standardized assessment frameworks (Davey, 2017) and the fact that they often disregard the specificities of the institutions analyzed – is common in the literature (Razak *et al.*, 2013) and supported by the results of this study.

In our view, this point is dichotomous because, as the objective results of the rankings come across subjective realities of HEIs, for different reasons (Lauder *et al.*, 2015), they should consider that their contributions to SD will occur in different ways. However, this perspective should not be viewed as a requirement for higher or lower scores, elevation or demotion. Such a situation implies that a balance in weighting needs to be found, seeking to avoid divergent metrics when expecting a fair, balanced and egalitarian assessment process and to some extent, predicting the contextual and structural specificities that permeate HEIs, globally.

Overall, the analysis based on the BP, demonstrated that both rankings were above average. Although not being an excellent outcome for both, this can be viewed with optimism as it demonstrates the supremacy of university rankings, from the perspective of the BP. As Shi and Lai (2013) stated, assessing the sustainability of an HEI is a multi-faceted problem. With this study, we intended to contribute substance and critical debate to this recent and still under-researched field of knowledge. Besides knowing what universities are doing or how they are addressing the SD challenges, we highlight the importance not only of evaluating the impacts of this movement (Findler *et al.*, 2019) but also the relevance of the tools for doing so, through clarified and scrutinized assessment systems.

The importance of sustainability-related university rankings goes beyond classificatory boundaries. It is inserted in a space dedicated to increasing commitment and constant search for the implementation of emphatic actions of HEIs toward SD issues. The contributions of such institutions, both necessary and urgent, are accompanied by the evolution of a society that is increasingly more attentive to its actual role in tackling social and environmental issues. In this scenario, rankings can be seen as information and disclosure sources for communication and consultation by society. Furthermore, beyond this, they can be understood as change guides and conductors, as they demonstrate how HEIs contribute and how they better do it in the view of a more balanced and sustainable future.

Concerning the SDGs, the involvement of universities with rankings that consider actions capable of helping to achieve the goals, provides different implications. In the short-term, university engagement can be seen as crucial to propagate the purpose and ideas of the 2030 Agenda in the daily lives of its stakeholders such as students, teachers and civil society. In the long run, as universities are responsible for training professionals of the future, once addressing and stimulating actions toward the SDGs, these future professionals will be able to foster the development of new sustainability practices, increasing the possibility that the goals stipulated in 2030 Agenda are met effectively.

We agree with Lauder *et al.* (2015) that, despite criticisms, sustainability rankings are here to stay. Overcoming such limitations and advancing the development and application of university sustainability rankings is a complex task and requires a participatory approach involving the university community stakeholders (Davey, 2017). Advances toward improving the rankings can ensure better universities' sustainability positive impacts and, consequently, a genuine contribution to society (Lauder *et al.*, 2015). Thus, considering the call for sustainability practices at universities to be developed in an integrated manner at all levels of the campus (Blanco-Portela *et al.*, 2018. Hooey *et al.*, 2017), the diversity of elements addressed in the rankings can function as a guide so that environmental and social actions are instilled in the routine of the institutions.

We are facing times in which universities are no longer considered excellent solely for their high scores, patents or number of scientific paper publications. HEIs are now charged with evincing their positive and negative impacts on SD and this – even timidly – is being reflected in the international university assessment, incorporated by global ranking systems. This movement rekindles the age-old debate as follows: In the end, what makes

“the best university?” Is it one that publishes more papers and has high citation scores or the one that positively impacts society? We recognize that this is a recent and not exhaustible debate, with multiple variables and contexts that should be considered. Within that ensemble, the university rankings and more specifically, the sustainability university rankings, must have their space.

Conclusions

This study aimed to answer the following question: How to evaluate the structure of global university sustainability rankings according to the BP framework? From a critical analysis of both rankings, UI-GM and THE-WUR, evidence has led us to two main conclusions. First, although the BP allows a comparative content assessment and aspects to be considered in the benchmarking of universal rankings, there is still a gap to be filled regarding the specificity of global university sustainability rankings. Second, although THE-WUR has performed better in this research, there are several opportunities for improvement, especially regarding the methodological procedures used in the assessment. We also point out that it may not be possible to arrive at a single and more appropriate ranking, but one that is more appropriate to a given temporal and contextual reality.

We understand that the main lesson of this study is related to the challenge of assessing sustainability in universities through rankings or other evaluation systems. It is a complex task comprising specific difficulties, such as understanding the sustainability concept *vis-à-vis* different institutions, the inequitable access to resources, the existence of cultural differences and distinct institutional environmental among the universities, globally. Thus, we comprehend that such difficulties cannot be reduced by a simple and decontextualized comparison of indicators.

The evidence presented and discussed in this paper can contribute to the growing knowledge on university sustainability rankings, shedding light on their required metrics, methodology organization and reporting. Although one can find in the literature a reasonable amount of studies regarding the UI-GM (Drahein *et al.*, 2019; Lauder *et al.*, 2015; Suwartha and Sari, 2013), maybe because of its recency, none could be found dedicated to the THE-WUR. This paper is the first one to compare these two ranking systems to the best of our knowledge. The literature points out that the sustainability-related university rankings are controversial due to the lack of clarity of some aspects (Shi and Lai, 2013). We then contribute to advancing on knowledge construction, generating new questions presented in the next topic.

As practical implications, we expect that the results in this study can be useful for university managers when deliberating on the most appropriate ranking for their institutions, considering the alignment of values and practices. In addition to being powerful instruments in helping universities toward SD, these rankings can be instruments of comparisons of the HEIs' trajectories and among competitors and, therefore, can be a source of support in the decision-making process. This study can also help HEIs better prepare for participating in sustainability-related rankings, collecting and consolidating pertinent data and increasing the process' efficiency.

By elucidating distinct and complementary aspects of these two worldwide recognized assessment frameworks, we hope to shed some light on the challenges and point out the opportunities for improvement on the existing university sustainability rankings or even give insights for the development of new ones. Also, in a provocative but constructive position concerning the regular HEIs rankings, we question their bases and understandings of “the best university,” as posited by rankings.

Limitations and future studies

This study has some limitations. First, we are aware that the UI-GM and THE-WUR are different from their own structure and methodology, given that they are built on distinct assumptions. This fact alone limits a strict comparison. However, it should not be an impeditive for content and critical scientific analysis, as the one we conduct in this paper. Second, the available data concerning both rankings were not as complete as necessary for a fair assessment considering the BP or the scale proposed by [Stolz et al. \(2010\)](#). The analysis on the BP may have been biased by the researchers, although an objective, impartial and consensual discussion led to the results found.

From the discussions and limitations, we point out some directions for future research, pointing out some research questions as follows:

- RQ1. What are the motivations for HEI to participate in sustainability rankings?
- RQ2. What are the institutional factors behind this movement and how do they influence HEI actions?
- RQ3. Do or will HEIs tend to look increasingly similar in their sustainability politics, programs and practices due to evaluations?
- RQ4. What are the benefits and disadvantages of participating in more than one ranking that intends to measure the same topic – sustainability performance?
- RQ5. What are the implications and lessons learnt for sustainability practices at the campus?

To answer some of these questions, theories related to institutionalism and organizational strategy could be of value, as well as accessing primary data, like interviews, as it could enrich the results.

Further, we suggest more comprehensive analytical studies, including local rankings, as the STARS, AISHE, ARISE and the People and Planet's University League. Considering that the rankings are distinct per nature and as it was not possible to use all the scales of assessment based on the BP, the same way [Stoltz et al. \(2010\)](#) did, this paper gives the opportunity for conducting the suggested analysis by elaborating a specific assessment framework for university sustainability rankings.

Finally, as we are focusing on global rankings, we recommend studies on the impact of universities considering the sustainability indices of their respective countries. Future research studies could focus on understanding the main difficulties in implementing the practices required by the global or local rankings, which would assist in the development of more contextualized and effective indicators for sustainability rankings. Also, analyzing the ways in which rankings indicators are met by participating universities in different countries could contribute to understanding how those institutions, inserted in different cultures, are adapting to the requirements of rankings. This could be done, for example, by investigating the best HEIs classified on the THE-WUR and the achievement of SGDs in the country level and/or the pertinent categories consisting of the UI-GM.

References

- Albareda-Tiana, S., Vidal-Raméntol, S. and Fernández-Morilla, M. (2018), "Implementing the sustainable development goals at university level", *International Journal of Sustainability in Higher Education*, Vol. 19 No. 3, pp. 473-497.
- Barth, M., Godemann, J., Rieckmann, M. and Stoltenberg, U. (2007), "Developing key competencies for sustainable development in higher education", *International Journal of Sustainability in Higher Education*, Vol. 8 No. 4, pp. 416-430.

- Blanco-Portela, N., Pertierra, L., Benayas, J. and Lozano, R. (2018), "Sustainability leaders' perceptions on the drivers for and the barriers to the integration of sustainability in Latin American higher education institutions", *Sustainability (Switzerland)*, Vol. 10 No. 8.
- Bizerril, M.X.A., Rosa, M.J. and Carvalho, T. (2018), "Construindo uma universidade sustentável: uma discussão baseada no caso de uma universidade portuguesa", *Avaliação*, Vol. 23 No. 2, pp. 424-447.
- Boer, P. (2013), *Assessing Sustainability and Social Responsibility in Higher Education Assessment Frameworks Explained*, Springer International Publishing, United Kingdom. pp. 121-137.
- Bothwell, E. (2019a), "The University -IRs 2019: results announced", Times Higher Education, available at: www.timeshighereducation.com/news/university-impact-rankings-2019-results-announced (accessed 15 November 2019).
- Bothwell, E. (2019b), "These universities are making the most impact on society. World economic forum", Times Higher Education, available at: www.weforum.org/agenda/2019/05/the-university-impact-rankings-2019-results-announced (accessed 16 November 2019).
- Butt, L., More, E. and Avery, G.C. (2014), "The myth of the 'green student': student involvement in Australian university sustainability programmes", *Studies in Higher Education*, Vol. 39 No. 5, pp. 786-804.
- Cotton, D.R.E., Winter, J., Miller, W. and Dalla Valle, L. (2018), "Is students' energy literacy related to their university's position in a sustainability ranking?", *Environmental Education Research*, Vol. 24 No. 11, pp. 1611-1626.
- Davey, E. (2017), "Recapturing the learning opportunities of university sustainability indicators", *Journal of Environmental Studies and Sciences*, Vol. 7 No. 4, pp. 540-549.
- Diez-Cañamero, B., Bishara, T., Otegi-Olaso, J.R., Minguez, R. and Fernández, J.M. (2020), "Measurement of corporate social responsibility: a review of corporate sustainability indexes, rankings and ratings", *Sustainability (Switzerland)*, Vol. 12 No. 5, pp. 21-53.
- Drahein, A.D., De Lima, E.P. and Da Costa, S.E.G. (2019), "Sustainability assessment of the service operations at seven higher education institutions in Brazil", *Journal of Cleaner Production*, Vol. 212, pp. 527-536.
- Farinha, C., Caeiro, S. and Azeiteiro, U. (2019), "Sustainability strategies in Portuguese higher education institutions: commitments and practices from internal insights", *Sustainability*, Vol. 11 No. 11.
- Findler, F., Schönherr, N., Lozano, R., Reider, D. and Martinuzzi, A. (2019), "The impacts of higher education institutions on sustainable development: a review and conceptualization", *International Journal of Sustainability in Higher Education*, Vol. 20 No. 1, pp. 23-38.
- Hazelkorn, E. (2014), "Reflections on a decade of global rankings: what we've learned and outstanding issues", *European Journal of Education*, Vol. 49 No. 1, pp. 12-28.
- Hooey, C., Mason, A. and Triplett, J. (2017), "Beyond greening: challenges to adopting sustainability in institutions of higher education", *Midwest Quarterly*, Vol. 58 No. 3, pp. 280-291.
- Huang, H.M. (2011), "A comparison of three major academic rankings for world universities: from a research evaluation perspective", *Journal of Library and Information Studies*, Vol. 9, pp. 1-25.
- Ireg (2006), *Berlin Principles on Ranking of Higher Education Institutions*, International Ranking Expert Group, Berlin.
- Lauder, A., Sari, R.F., Suwartha, N. and Tjahjono, G. (2015), "Critical review of a global campus sustainability ranking: Greenmetric", *Journal of Cleaner Production*, Vol. 108, pp. 852-863.
- Lazzarini, B. and Pérez-Foguet, A. (2018), "A. Profiling research of the engineering academics who successfully promote education in sustainable human development", *Journal of Cleaner Production*, Vol. 172, pp. 4239-4253.
- Leal Filho, W. (2011), "About the role of universities and their contribution to sustainable development", *Higher Education Policy*, Vol. 24 No. 4, pp. 427-438.

- Leal Filho, W., Brandli, L.L., Salvia, A.L., Rayman-Bacchus, L. and Platje, J. (2020), "COVID-19 and the UN sustainable development goals: threat to solidarity or an opportunity?", *Sustainability (Switzerland)*, Vol. 12 No. 13, pp. 1-14.
- Liu, Z., Moshi, G.J. and Awuor, C.M. (2019), "Sustainability and indicators of newly formed world-class universities (NFWCUs) between 2010 and 2018: empirical analysis from the rankings of ARWU, QSWUR and THEWUR", *Sustainability*, Vol. 11 No. 10, pp. 2745-2774.
- Lozano, R. (2006), "A tool for a graphical assessment of sustainability in universities (GASU)", *Journal of Cleaner Production*, Vol. 14 No. 9-11, pp. 963-972.
- Lozano, R., Lukman, R., Lozano, F.J., Huisingh, D. and Lambrechts, W. (2013), "Declarations for sustainability in higher education: becoming better leaders, through addressing the university system", *Journal of Cleaner Production*, Vol. 48, pp. 10-19.
- Lozano, R., et al. (2015), "A review of commitment and implementation of sustainable development in higher education: results from a worldwide survey", *Journal of Cleaner Production*, Vol. 108, pp. 1-18.
- Lukman, R., Krajnc, D. and Glavič, P. (2010), "University ranking using research, educational and environmental indicators", *Journal of Cleaner Production*, Vol. 18 No. 7, pp. 619-628.
- Marginson, S. and Van Der Wende, M. (2007), "To rank or to be ranked: the impact of global rankings in higher education", *Journal of Studies in International Education*, Vol. 11 Nos 3/4, pp. 306-329.
- ONU (2015), "Transformando Nosso Mundo: A Agenda 2030 para o Desenvolvimento Sustentável", available at: <https://nacoesunidas.org/pos2015/agenda2030/> (accessed 16 November 2019).
- Puertas, R. and Martí, L. (2019), "Sustainability in universities: DEA-GreenMetric", *Sustainability*, Vol. 11 No. 14, pp. 3766-3784.
- Ragazzi, M. and Ghidini, F. (2017), "Environmental sustainability of universities: critical analysis of a green ranking", *Energy Procedia*, Vol. 119, pp. 111-120.
- Ramos, T.B., Caeiro, S., Van Hoof, B., Lozano, R., Huisingh, D. and Ceulemans, K. (2015), "Experiences from the implementation of sustainable development in higher education institutions: environmental management for sustainable universities", *Journal of Cleaner Production*, Vol. 106, pp. 3-10.
- Razak, D.A., Sanusi, Z.A., Jegatesen, G. and Khelghat-Doost, H. (2013), "Alternative university appraisal (AUA): reconstructing universities' ranking and rating toward a sustainable future", *Sustainability Assessment Tools in Higher Education Institutions*, pp. 139-154.
- Stolz, I., Hendel, D.D. and Horn, A.S. (2010), "Ranking of rankings: benchmarking twenty-five higher education ranking systems in Europe", *Higher Education*, Vol. 65 No. 5, pp. 507-505.
- Shi, H. and Lai, E. (2013), "An alternative university sustainability rating framework with a structured criteria tree", *Journal of Cleaner Production*, Vol. 61, pp. 59-69.
- Shriberg, M. (2002), "Institutional assessment tools for sustainability in higher education: strengths, weaknesses, and implications for practice and theory", *Higher Education Policy*, Vol. 15 No. 2, pp. 153-167.
- Sonetti, G., Lombardi, P. and Chelleri, L. (2016), "True green and sustainable university campuses? Toward a clusters approach", *Sustainability (Switzerland)*, Vol. 8 No. 1, pp. 1-23.
- Stolz, I., Hendel, D.D. and Horn, A.S. (2010), "Ranking of rankings: benchmarking twenty five higher education ranking systems in Europe", *Higher Education*, Vol. 60 No. 5, pp. 507-528.
- Storey, M., Killian, S. and O'regan, P. (2017), "Responsible management education: mapping the field in the context of the SDGs", *International Journal of Management Education*, Vol. 15 No. 2, pp. 93-103.
- Suwartha, N. and Sari, R.F. (2013), "Evaluating UI greenmetric as a tool to support green universities development: assessment of the year 2011 ranking", *Journal of Cleaner Production*, Vol. 61, pp. 46-53.
- Taylor, P. and Braddock, R. (2007), "International university ranking systems and the idea of university excellence", *Journal of Higher Education Policy and Management*, Vol. 29 No. 3, pp. 245-260.

- THE World University Rankings (2019a), "The University -IRs 2019: methodology", London, available at: www.timeshighereducation.com/world-university-rankings/methodology-impact-rankings-2019 (accessed 15 November 2019).
- THE World University Rankings (2019b), University -IRs 2019, London, available at: www.timeshighereducation.com/rankings/impact/2019/overall#!/page/0/length/25/sort_by/rank/sort_order/asc/cols/undefined (accessed 15 November 2019).
- THE World University Rankings (2019c), "User guide", Version 1.5.1, London, October 2019.
- Trowler, P., Hopkinson, P. and Boyes, L. (2013), "Comerford. Institutional change towards a sustainability agenda: how far can theory assist?", *Tertiary Education and Management*, Vol. 19 No. 3, pp. 267-279.
- UI Green Metric World University Rankings (2019a), "Overall Rankings", available at: <http://greenmetric.ui.ac.id/overall-rankings-2019/> (accessed 12 December 2019).
- UI Green Metric World University Rankings (2019b), "Guideline", available at: <https://greenmetric.ui.ac.id/guidelines-page/> (accessed 12 December 2019).
- Vargas, V.R., Lawthom, R., Prowse, A., Randles, S. and Tzoulas, K. (2019), "Sustainable development stakeholder networks for organizational change in higher education institutions: a case study from the UK", *Journal of Cleaner Production*, Vol. 208, pp. 470-478.
- Vos, R.O. (2007), "Defining sustainability: a conceptual orientation", *Journal of Chemical Technology and Biotechnology*, Vol. 82 No. 4, pp. 334-339.
- Webster, D.S. (1986), *Academic Quality Rankings of American Colleges and Universities*, Charles C. Thomas, Springfield, IL.
- Zamora-Polo, F., Sánchez-Martín, J. and Espejo-Antúnez, L. (2019), "What do university students know about sustainable development goals? A realistic approach to the reception of this UN program amongst the youth population", *Sustainability*, Vol. 11 No. 13, pp. 3533-3552.
- Zutshi, A., Credo, A. and Connelly, L. (2019), "Education for sustainable development: emerging themes from adopters of a declaration", *Sustainability*, Vol. 11 No. 1.

Further reading

- Leal Filho, W., *et al.* (2019), "The role of green and sustainability offices in fostering sustainability efforts at higher education institutions", *Journal of Cleaner Production*, Vol. 232, pp. 1394-1401.

Corresponding author

Barbara Galleli can be contacted at: barbara.galleli@ufpr.br

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgrouppublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com